On Positive Integers Satisfying $n = s(n) \cdot \tau(n)$: The Anand Numbers

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Abstract

We classify positive integers n satisfying $n = s(n) \cdot \tau(n)$, where $s(n)$ is the sum of decimal digits (base 10) and $\tau(n)$ is the number of positive divisors (OEIS A000005). We prove that exactly three numbers satisfy this for $n \geq 1$: $1, 80, 108$. We term such integers **Anand Numbers**. The proof uses $\tau(n) \leq 2\sqrt{n}$ and $s(n) \leq 9k$ for k-digit numbers, giving an upper bound $n \leq 5184$, verified computationally up to 10^6. $\tau(0)$ is undefined (0 has infinitely many divisors), hence 0 is excluded. The triple is composed solely of tetradic digits $\{0, 1, 8\}$, and exhibits Harshad (A005349), refactorable (A033950), and hyperfactorial $H(3) = 108$ properties. No exact sequence for this property was found in OEIS, MathWorld, or literature as of 25 Oct 2026.

1 Introduction and Definitions

Let $n \in \mathbb{Z}^+$ be a positive integer. We investigate the interaction between the additive digit-sum function and the multiplicative divisor function.

Definition 1 (Anand Numbers)**.** We define the following components:

1. $s(n)$: The sum of decimal digits of n in base 10.
2. $\tau(n)$: The number of positive divisors of n (OEIS A000005), defined for $n \geq 1$ only. Note that $\tau(0)$ is undefined as 0 has infinitely many divisors, hence $n = 0$ is excluded.
3. A positive integer $n \geq 1$ is termed an **Anand Number** if it satisfies:

$$n = s(n) \cdot \tau(n) \tag{1}$$

2 Main Theorem and Proof

Theorem 1. *The equation $n = s(n) \cdot \tau(n)$ has exactly three solutions in positive integers $n \geq 1$:*

$$n \in \{1, 80, 108\}$$

Proof. Let n be a k-digit positive integer in base 10. Thus, $n \geq 10^{k-1}$. We utilize classical standard upper bounds:

1. $s(n) \leq 9k$
2. $\tau(n) \leq 2\sqrt{n}$ (Burton, Elementary Number Theory)

Substituting these bounds into Equation (1) yields:

$$n \leq (9k) \cdot (2\sqrt{n}) = 18k\sqrt{n}$$

Dividing both sides by $\sqrt{n}$ and squaring gives:

$$\sqrt{n} \leq 18k \implies n \leq 324k^2 \tag{2}$$

Since $n \geq 10^{k-1}$, we obtain:

$$10^{k-1} \leq 324k^2$$

Evaluating this inequality for small integer values of k:

- For $k = 1$: $10^0 = 1 \leq 324(1)^2 = 324$ (Valid)
- For $k = 2$: $10^1 = 10 \leq 324(2)^2 = 1296$ (Valid)
- For $k = 3$: $10^2 = 100 \leq 324(3)^2 = 2916$ (Valid)
- For $k = 4$: $10^3 = 1000 \leq 324(4)^2 = 5184$ (Valid)
- For $k = 5$: $10^4 = 10000 > 324(5)^2 = 8100$ (**False**)

For all $k \geq 5$, $10^{k-1} > 324k^2$ holds strictly. Therefore, any solution must satisfy $k \leq 4$, establishing the absolute theoretical upper bound:

$$n \leq 5184$$

An exhaustive search over the domain $1 \leq n \leq 5184$ (and verified computationally up to 10^6 for safety) yields exactly three solutions: $\{1, 80, 108\}$. Furthermore, among numbers composed solely of tetradic digits $\{0, 1, 8\}$ up to 20736 (161 in count), only $1, 80$, and 108 satisfy the condition. □

3 Discussion and Properties

The triple $\mathcal{A} = \{1, 80, 108\}$ exhibits remarkable number-theoretic properties:

1. **Tetradic Digits:** All elements use exclusively the digit set $\{0, 1, 8\}$.
2. **Harshad Numbers (A005349):**

- $1 \pmod 1 = 0$ ($s(1) = 1$)
- $80 \pmod 8 = 0$ ($s(80) = 8$)
- $108 \pmod 9 = 0$ ($s(108) = 9$)

3. **Refactorable Numbers (A033950):**
   - $\tau(1) = 1 \mid 1$
   - $\tau(80) = 10 \mid 80$
   - $\tau(108) = 12 \mid 108$
4. **Hyperfactorial Connection:** $108 = 2^2 \cdot 3^3 = H(3)$, the third hyperfactorial.

4 Novelty Check

As of 25 Oct 2026, searches across OEIS, MathWorld, and mathematical literature for the property "sum of digits times number of divisors equals n" returned no prior matches or exact sequences. Thus, the classification of Anand Numbers is believed to be entirely new.

5 Conclusion and Future Work

We proved that $n = s(n) \cdot \tau(n)$ has exactly three positive solutions: $1, 80, 108$, termed **Anand Numbers**. Future work will explore base-b generalizations and functions of the form $s_b(n) \cdot \tau(n)$. Deeper structural invariances of this triple are reserved for subsequent publication and the author's forthcoming book *Manujatva, Mrityu and Moksha (Yoga Darshan)*.

References

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